

Introduction to Artificial Intelligence

Course Metadata

Credit Structure	L-T-P: 4-0-0 Total: 4
Offering Semester	3 rd Semester, Autumn
Level of Subject	UG Level
Compulsory or elective?	Core
Pre-requisite	Programming and Data Structures
Faculty Names	Koustav Rudra, Plaban Bhowmik, Arnab Sarkar, Sudeshna Sarkar
Overlapping Subjects	Artificial Intelligence (CS60045)~60%, Artificial Intelligence Foundation and Applications (AI61005)~60%
Justification of overlap	This course will be offered as a CORE to the BTech in AI program

Course Description

The primary objective of this course is to introduce the gamut of real-life problems where AI techniques can be successfully applied and detailing the necessary foundations to enable solving those practical problems.

Course Content

The course will revolve around several practical case-studies and provide foundational understanding to apply AI methods in solving them effectively. This course shall also provide necessary theoretical insights spanned across three layers – knowledge representation and logic; search and reasoning frameworks; and ramifications of AI techniques under various scenarios/constraints. These foundational and theoretical aspects will be enabled by suitable tools and appropriate communication interface designs, thereby bringing a holistic view of problem solving through applications of AI techniques. Following topics will be covered:

Introduction to AI, State-Space Search, Heuristic Search, Heuristic Search with Time and Memory Constraints, Reasoning & Deduction, Local Search, Handling Uncertainty/Incomplete Information, Fuzzy Logic, Bayesian Networks and Markov Decision Process, Problem Solving in Distributed, Multi-Agent Scenarios, Case Studies, Summary and Conclusions

Lecture-wise Breakup

Sl No.	Module	Unit	Module Hr (48)
1	Introduction to AI	Historical Evolution and AI today, Architecture of AI Systems (Theory: Knowledge, Logic, Search, Reasoning, Planning, Learning, Tools and Applications, Interface (Sensing, Actuation, Control), Domains and Scope of AI Applications (Interdisciplinary Problem Solving, Intelligent/Expert Knowledge based Systems, Case-studies and Success Stories)	3
2	State-Space Search	Search problem Representation, Uninformed Search Strategies (BFS, DFS, Uniform Cost, Iterative Deepening, Bi-directional etc.), Informed/Heuristic Search (DFBB, Best-first Search / A* Search, Search over Large Knowledge Graphs)	5
3	Constraint Satisfaction Problems (CSP)	Defining a problem as a CSP, Solving: Backtracking and CSP Search (Variable and Value Ordering, Forward Checking and Arc Consistency, Techniques for Exploiting Problem Structures (Tree-structured CSPs etc.))	5
4	Advanced Search Techniques	IDA* search, RBFS, Contract search, Anytime search (Memory-bounded search, Search with Real-time adjustments, Multi-objective Search), Local Search: Hill climbing, Simulated Annealing, Genetic Algorithms etc., Adversarial Search (Minimax, Alpha-Beta Pruning, Horizon Effect)	5
5	Reasoning & Deduction	Propositional and Predicate Logic, Resolution Refutation, Proof Techniques & Tools: SAT and SMT solvers	5
6	Reasoning under Uncertainty	Modeling Uncertainty with Probabilistic Reasoning, Bayesian Belief Networks, Conditional Independence and D-separation, Probabilistic Reasoning and inference	5
7	Fuzzy Logic	Fuzzy Set Representation, Membership Function, Linguistic Variable – Hedges, Fuzzy Relation, Fuzzy Rule, Set of Support, Fuzzy Arithmetic, Fuzzy Inference Engine	4
8	Planning	Introduction to Planning Problem, Planning under uncertainty (full/partial	6

		observability, deterministic / stochastic states, discrete / continuous periodicity), Markov Decision Process (MDP), Partially Observable Markov Decision Process (POMDP)	
9	Distributed, Multi-Agent Scenarios	Collaborative Problem Solving (Eg. Crowdsourced recommender), Multi-agent Path Finding, Conflict-Based Search, Competitive Problem Solving (Eg. Multi-player games), Utility based (dynamic) Problem Solving (Ex. Stock Market)	6
10	Case Studies	Transportation (Dynamic vehicle routing, pick-up and drop, On-demand high-capacity ride-sharing via dynamic assignment, Optimal Pricing for Improving Efficiency of Taxi Systems), Game Design and Play (Alpha-beta pruning in Chess, Alphago), VLSI System Design (Specification, Validation, Synthesis), Socially and Globally Relevant (Weather forecasting)	4

Books

1. **Artificial Intelligence: A Modern Approach**, Stuart J. Russell and Peter Norvig, 3rd Edition, Prentice Hall, 2009.
2. **Principles of Artificial Intelligence**, Nils J. Nilsson, Morgan Kaufmann, 1982.
3. **Artificial Intelligence: A New Synthesis**, Nils J. Nilsson, 1st Edition, Morgan Kaufmann, 1998.

Reference

1. **Heuristic Search: Theory and Applications**, Stefan Edelkamp and Stefan Schroedl, 1st Edition, Morgan Kaufmann, 2011.
2. **Heuristics: Intelligent Search Strategies for Computer Problem Solving**, Judea Pearl, Addison Wesley Longman Publishing Co., 1984.
3. **The Quest for Artificial Intelligence**, Nils J. Nilsson, Cambridge University Press, 2009.
4. **Logic in Computer Science: Modelling and Reasoning about Systems**, Michael Huth and Mark Ryan, 2nd Edition, Cambridge University Press, 2004.
5. **The Cambridge Handbook of Artificial Intelligence**, Keith Frankish and William M. Ramsey (editors), Cambridge University Press, 2014.
6. Relevant research papers